

$$\Phi(s) = \frac{25K_1}{s^2 + (0.8 + 25K_1K_t)s + 25K_1}$$



$$\text{target: } \omega_n = 6, \zeta = 1$$



$$25K_1 = \omega_n^2, \quad 0.8 + 25K_1K_t = 2\zeta\omega_n$$



$$K_1 = 1.44, \quad K_t = 0.31$$